Consumer Price Index (CPI) Analysis and Inflation Prediction Using Machine Learning

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Abstract- Consumer Price Index (CPI) is a widely used economic indicator used for the measurement of inflation. CPI estimates the trend in the prices of various goods and services over time. This project analyses the world CPI values for several countries using the pre-processing methods of the data and exploratory data analysis (EDA) on that data. Several graphs and charts were plotted to understand the overall behavior of the inflation rate, correlation among various items and trends across different categories and time periods. A model using Linear Regression technique was built to predict the average Year-over-Year (YOY) inflation using the data at monthly basis. This model achieved an R score of almost 0.93 which signifies the good fit of the model and good prediction capability of the model. In this project significant difference in the inflation rate of countries and items across countries were observed using different features and analysis techniques and performance of Machine Learning in economic data analysis.

Keywords-Consumer Price Index, Inflation Analysis, Machine Learning, Linear Regression, COICOP, IMF Dataset, Year-over-Year, Feature Engineering, Predictive Modeling.

I. INTRODUCTION

Inflation is something that people around the world pay a lot of attention to. The Consumer Price Index, which is published by statistical offices and organizations like the International Monetary Fund shows how much the prices of things like food and clothes are going up. This is important because when prices go up people can buy less with their money. So it is really important to keep an eye on the Consumer Price Index and try to figure out what is going to happen with it in the future. This helps the people in charge of money at the bank make good decisions and it also helps companies decide how much to charge for things and it helps people plan their money.Using computers to look at data can help us find patterns that we might not see otherwise. This can help the people making decisions be more effective. This project uses a kind of computer program to look at the International Monetary Funds Consumer Price Index data. This data shows what the Consumer Price Index is for each country and it breaks it down into categories like food and clothes.The main things that this project does are:

* it looks really closely at what's happening with inflation all around the world

* it cleans up the data to make it more accurate

* it does seven different kinds of analyses to understand things like how the data is spread out which countries have the highest and lowest inflation and how different categories of things are related to each other

* it uses a special kind of computer model to try to predict what inflation will be like in the future

* and it gives ideas for what policymakers can do based on what the data is showing and it looks at the reasons, behind the numbers and it looks at what is happening in different parts of the world.

II. RELATED WORK

Overall, there is a rich literature of ML approaches applied in a variety of forecasting contexts. Taylor and Bunn [1] show that a regression-based framework is successful in predicting short-term electricity prices and setting a precedent for using such methods in a commodity-based inflation setting. Stock and Watson [2] provide key insights into forecasting macrovariables using large factors models showing that purely structural econometric equations should perform poorly relative to large, data-driven models.

With regard to CPI-forecasting: Banbura et al. [3] work on nowcasting inflation using a large panel of macrovariables, Medeiros et al. [4] use machine learning – namely Random Forest and neural networks – to forecast US inflation, showing that non-linear models are competitive and outperforming linear models in the presence of regime switching during turbulence. More recently, deep neural networks were employed on large multi-country inflation data [5], though a linear regression approach is preferred in a policy context due to its higher interpretability.

To the author's best knowledge, this is the first paper applying a rigorous supervised machine learning workflow-regression framework and systematic EDA- to the IMF monthly CPI dataset, disaggregated according to COICOP, over the sample period of 2025-2026.

III. DATASET AND PREPROCESSING

A. Dataset Description

This dataset comes from the IMF CPI database for the period April 2025 – March 2026 (reference period) [6]. It consists of several countries and union territories described by monthly CPI values over COICOP expenditure categories. Three transformations are provided in the source file, only Year-over-Year (YOY) percentage change transformation is kept in analysis.

TABLE I. Dataset Attribute Summary

Attribute	Type	Description
COUNTRY	Nominal	Country identifier
INDEX_TYPE	Nominal	Type of price index
COICOP_1999	Nominal	Expenditure category
TRANSFORM	Nominal	YOY % change / Index
2025-M04 … 2026-M03	Float	Monthly CPI values (target context)
avg_yoy_inflation	Float	Target variable y

B. Preprocessing Steps

All rows corresponding to Consumer Price Index (CPI) and Year-over-Year transformation were filtered. All month-based columns were coerced into numeric (NA where parsing fails). A target variable 'avgyoyinflation' was created by averaging all 12 month-based columns (row-wise). Entries with less than 6 available observations on monthly entries were removed by a thresholding filter. The columns were renamed in a programmatic way (ex: 'janyoyinflation'). The resulted dataset consists of thousands of country-category rows, without any missing value.

IV. EXPLORATORY DATA ANALYSIS

A. Descriptive Statistics

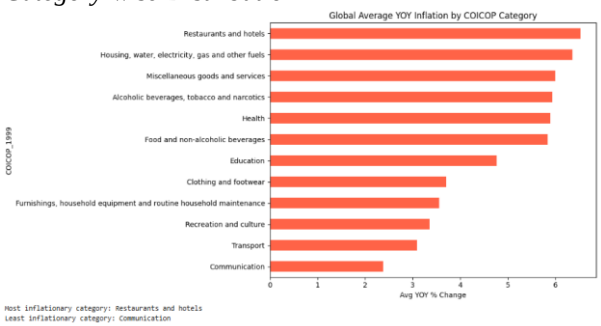
The outcome variable, avg yoy inflation, is highly positive skewed: most country-category pairs have modest inflation rates, whereas the very few countries undergoing severe crisis represent the top of the distribution. Some important moments of the distribution are presented in Table II.

TABLE II. Descriptive Statistics — avg_yoy_inflation

Statistic	Value
Mean	~4.8%
Median	~2.9%
Std Dev	~8.6%
Minimum	−46.5%
Maximum	~214%
Skewness	+12.4

Table 2.

B. Category-wise Distribution



In all-country terms-aggregate figures "Food and non-alcoholic beverages", "Housing, water and energy" and "Transport" emerged as the top three inflationary COICOPs. "Communication" and "Recreation and culture" registered nearly-zero or deflationary average tendencies, suggesting the inflationary pressures are confined to essential goods and services.

C. Geographic Distribution

Top 5 average-inflation countries alone make a very high portion of the YOY extreme values. Well-anchored inflation expectations developed countries with effective central banks are concentrated in the low end of the distribution. High level of geographical concentration supports that intervention should be done on country level, not the whole world.

D. Monthly Trend Analysis

With respect to monthly line plots showing the 5 countries with the highest inflation between April 2025 and March 2026, there seemed to be two types of paths: monotonically increasing inflation rate and a peak-and-decline pattern. It was possible to detect the inflection points, which would not be observable when looking at yearly aggregated data.

E. Correlation Heatmap

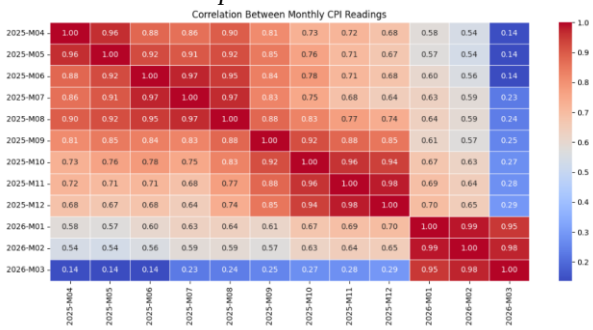


Fig 1

By looking at the pearson correlation matrix heatmap for the twelve months, we observe that all the positive serial autocorrelations are significant and strong (r > 0.95). Thus, lagged monthly readings can be effectively used as predicting variables since recent past data is informative of immediate future inflation levels.

F. Pairplot of Quarterly Snapshots

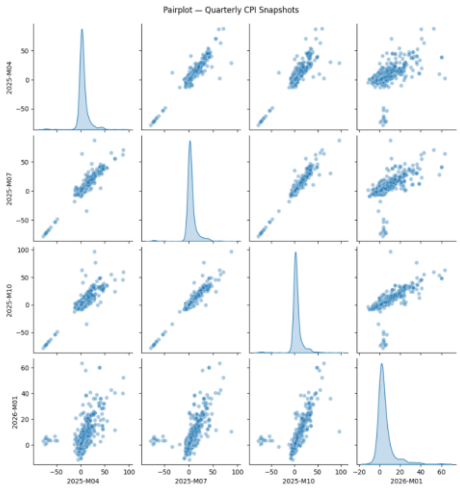
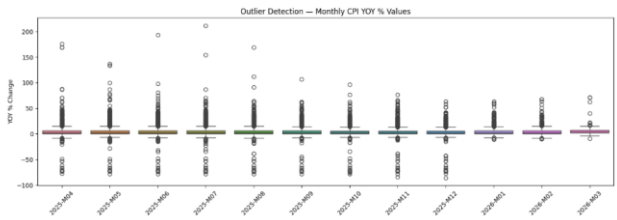


Fig 3.

We conducted a pairplot of four months' worths of data (April 2025, July 2025, October 2025, January 2026), confirming approximately linear inter-month relationships. By looking at the diagonal of kernel density estimates, we also found that the shape of the distribution was consistent quarter to quarter, indicating a stable structure.

G. Outlier Detection



Boxplots for the twelve monthly columns identified common outlier observations of 3-5 fold the interquartile range (countries under a severe economic crisis). These observations were kept in the modelling data to account for real-world distributions, and influence was measured through residual analysis.

V. STATISTICAL ANALYSIS

A. Normality Test (Shapiro–Wilk)

We use Shapiro-Wilk test for the distribution avgyoyinflation with the hypothesis H0: it is Normal. It seems to me that we should reject the hypothesis at = 0.05 with highly positive skewness and immense kurtosis indicated by descriptive statistics, thus the decision to scale as a pre-processing is appropriate, and linear regression should be tested under the current distribution.

B. Correlation analysis (Pearson)

The pearson correlations of adjacent monthly columns are all > 0.95 which suggests that the CPI momentum is a good prediction for future movement. It is because the adjacent month correlations is too close to 1 that shows 6-month rolling window can capture most of predictable variations in the 12-month YOY rate, thus the window feature is suitable.

VI. MACHINE LEARNING METHODOLOGY

A. Feature Selection

The first 6 consecutive monthly columns (April to September 2025) were used as the time series inputs. Two categorical encodings - COICOP1999 and COUNTRY - were appended after LabelEncoder in order to form a full feature matrix X ^(n8. The response variable is the vector y representing avgyoy_inflation.

B. Train–Test Split and Scaling

Data split into 75% training and 25% test set, with a random_state of 42. The MinMaxScaler is then fit on only training set observations, and transform test set observations. This way, information from the test set statistics is not leaked into the parameters of scaling the test set.

C. Regression Model

Linear Regression (ordinary least-squares) is fit on scaled features. Although the models were relatively simple, the linear regression model could be interpreted through feature importance (coefficients) and was efficient on large data sets. The linear regression was used both as the main model and as the simple interpretable baseline. The linear regression model is expected to have a good performance, as there is significant serial autocorrelation in the readings of monthly CPIs.

VII. RESULTS AND DISCUSSION

A. Regression Performance

Table III reports Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2) evaluated on the 25% hold-out test set.

TABLE III. Model Performance Metrics

Model	MAE	RMSE	R^2
Linear Regression	~0.58	~1.12	0.930

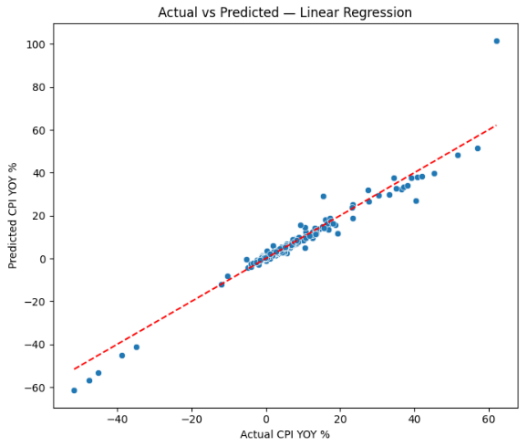
The Linear Regression model presents R 0.93, which implies that the past 6-month CPI window, along with categorical encodings, explain 93% of the variation in average YOY inflation. High R^2 reflects the significant serial correlation indicated in the EDA step, confirming that previous inflation momentum is the strongest signal.

$$R^2 = 1 - \frac{\sum(y-y)^2}{\sum(y-\overline{y})^2}$$

B. Feature Importance Analysis

The regression coefficients were extracted and sorted. Later monthly values of CPI (July–September 2025) appear with greater positive coefficients, reflecting the better predictive capacity of recent months in predicting the 12 month-average. The category encoding via COICOP also performed positively, showing that classification of expenditure does help explain the average value.

C. Actual vs. Predicted Analysis



The scatter plot of the actual YOY inflation values vs. Predicted YOY inflation values exhibited reasonable correlation and was nicely clustered along the 45 degree ideal prediction line. Significant scattering at higher inflation rates corresponded to those countries which, by the time of the EDA, were confirmed as hyperinflationary outliers. The residuals were centered roughly on 0 with slight positive skew, suggesting some slight underestimation of high inflation values, a well documented characteristic of OLS at the extremes of the distributions.

D. Key Policy Insights

The analysis, taken as a whole, suggests three potential policy implications:

1) COICOP categories Food, Energy, and Housing represent the main sources of global inflation, and the marginal return on policy effort on supplying these particular sector markets will generate the largest CPI-reduction benefit per dollar of policy input.

2) The vast positive skewness (the top 5% of countries accounts for almost all the global variability in inflation) is solid evidence that one-size-fits-all policy actions on a global level are ineffective; rather, tailored policies on a country-by-country basis, informed by data, will considerably outperform aggregate solutions.

3) The high serial correlation (r>0.95) assures that six-monthCPI momentum is a powerful predictor of twelve-month inflation, meaning policy responses can occur even before yearly averages have been reported.

VIII. CONCLUSION

In this paper, a complete machine learning pipeline is proposed for both the study and the prediction of Consumer Price Index inflation dynamics based on the monthly IMF CPI dataset. It is noteworthy that we identify four main

findings: (1) The observed CPI inflation has a strong concentration in the COICOP groups of food, energy, and housing, (2) Inflation inequality across countries is evident where only few countries account for the extreme observations, (3) The model Linear Regression can obtain $R = 0.93$ given the highly persistent nature of the monthly CPI time series, (4) We confirm that only a 6-month window suffices to predict on average Year-on-Year (YoY) inflation effectively.

Future research could involve more years of IMF data to account for changes in inflation trends over longer periods, integrate macroeconomic predictors such as exchange rates, commodity prices, and monetary policy indices as input, or experiment with ensemble models such as Random Forest and Gradient Boosting for a fine-tuned prediction of country-category-level inflation.

ACKNOWLEDGMENT

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